PROJECT CASE STUDY | SOFTWARE QUALITY ENGINEERING

Healthcare & Medical Imaging Platform

End-to-End Multi-Platform Quality Assurance, Interoperability & Regulatory Validation

INDUSTRY	DOMAIN	ENGAGEMENT	PLATFORMS
Healthcare & Medical Technology	Medical Imaging & Diagnostics	Software Quality Engineering & Testing	Web, iOS, Android, iPad, Windows, Mac & APIs

PROJECT OVERVIEW

We delivered comprehensive Software Quality Engineering and Testing services for an enterprise healthcare imaging and diagnostic platform. Designed to power critical clinical operations, the platform orchestrates diagnostic imaging workflows, patient record management, clinical reporting, and the secure exchange of healthcare information across diverse medical environments.

Our engagement ensured seamless interoperability across connected medical devices, Electronic Health Record (EHR) systems, and multi-platform applications—guaranteeing high reliability, strict data integrity, and uninterrupted clinical operations.

QUALITY ENGINEERING & TESTING SCOPE

Our QA team executed a full-lifecycle testing strategy spanning web, mobile, tablet, desktop, API, and connected-device ecosystems:

- **Multi-Platform Application Testing:** Executed functional, regression, integration, smoke, sanity, and usability testing across Web, iOS, Android, iPad, Windows, and macOS applications.
- **Clinical & Imaging Workflow Validation:** Verified end-to-end patient assignment flows, diagnostic reporting accuracy, and medical imaging data and metadata integrity.
- **Healthcare System Interoperability:** Validated seamless data synchronization across EHR systems, connected diagnostic medical devices, and external clinical platforms.
- **API & Data Security Validation:** Performed comprehensive REST API request/response validation, ensuring secure data transmission and offline/online mobile synchronization.
- **Performance & Localization Testing:** Conducted high-volume workflow stress testing under clinical load alongside complete UI/UX localization validation for English and Spanish.
- **Agile QA & Defect Governance:** Managed end-to-end test cases, traceability, and defect lifecycles in close collaboration with product and engineering teams.

Contact our team at **contact@horigineitsolutions.com** or visit **www.horigineitsolutions.com**to discuss your project requirements.

AUTOMATION & TESTING TECHNOLOGY STACK

Test Automation	Selenium, Appium, TestNG, JUnit, Cucumber BDD
API & Integration Testing	Postman, REST Assured, Swagger / OpenAPI
Test & Defect Management	Jira, Jama
Reporting & Analytics	Allure, ExtentReports
Target Platforms	Web Browsers, iOS, iPadOS, Android, Windows, macOS

HEALTHCARE DOMAIN & COMPLIANCE FOCUS

- **Clinical & Diagnostic Domain Coverage:** Extensive validation across medical imaging diagnostics, clinical operations, telemedicine workflows, healthcare analytics, and patient data management.
- **Regulatory & Privacy Compliance:** All testing activities were structured around applicable healthcare security, data privacy, patient safety, and regulatory quality standards to safeguard sensitive diagnostic and patient records.

KEY BUSINESS IMPACT

- **Reliable Clinical Workflows:** Ensured uninterrupted diagnostic imaging and clinical operations across hospital and remote environments.
- **100% Data Accuracy & Integrity:** Safeguarded sensitive imaging metadata and patient records across high-volume transfers and EHR integrations.
- **Consistent Multi-Device Experience:** Delivered uniform stability and usability across desktop, web, tablet, and mobile endpoints.
- **Accelerated Release Confidence:** Enabled early defect identification in Agile sprints, reducing production risks and strengthening release readiness.

Note: Client identity and proprietary system names have been anonymized to protect confidentiality. This is for informational purposes only. For medical advice or diagnosis, consult a professional.

Contact our team at **contact@horigineitsolutions.com** or visit **www.horigineitsolutions.com**to discuss your project requirements.